# Abstract Algebra

SF1678

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# 1. Sets

## 1.1. RELATIONS

### 1.1.1. Set Theory

**Definition 1.1 (Tuple)**

A tuple (or sometimes **list**) is a finite, ordered collection of elements, possibly with repetitions.

An $n$-tuple is of length $n$. A 2-tuple is called an **ordered pair**.

### 1.1.2. Morphisms

**Definition 1.2 (Map)**

Subsets of the Cartesian product $A \times B$ are called **relations**. A **map** is a **well-defined** relation written

$$f \subseteq A \times B \quad \text{or} \quad f : A \to B \quad \text{or} \quad A \xrightarrow{f} B$$

and is

- **surjective** or **onto** if $f(A) = B$, which is equivalent to being **right invertible** with $f \circ f^{-1} = \mathrm{id}_B$.
- **injective** or **one-to-one** if $a_1 \neq a_2 \implies f(a_1) \neq f(a_2)$, which is equivalent to being **left invertible** with $f^{-1} \circ f = \mathrm{id}_A$.
- **bijective** or a **one-to-one correspondence** if both above apply, which is equivalent to being **invertible**.

**Definition 1.3 (Homomorphism)**

A **homomorphism** is a map between algebraic structures that preserves the structure.

A group homomorphism $\varphi : G \to H$ satisfies

$$\varphi(g * \gamma) = \varphi(g) \cdot \varphi(\gamma)$$

for all $g, \gamma \in G$.

**Theorem 1.1**

If $\varphi : G \to H$ is a group homomorphism,
(i) $\varphi(\mathrm{id}_G) = \mathrm{id}_H$
(ii) $\varphi(g^{-1}) = \varphi(g)^{-1}$ for all $g \in G$

**Definition 1.4 (Kernel)**

The **kernel** of a group homorphism $\varphi : G \to H$ is the set

$$\ker \varphi = \{g \in G \mid \varphi(g) = \mathrm{id}_H\}.$$

**Definition 1.5 (Isomorphism)**

An **isomorphism** is a bijective homomorphism.

If there exists an isomorphism between two algebraic structures, they are said to be **isomorphic**, denoted with ≅.

**Definition 1.6 (Endomorphism)**

An **endomorphism** is a homomorphism from a mathematical object to itself.

**Definition 1.7 (Automorphism)**

An **automorphism** is simultaneously an isomorphism and an endomorphism: a bijective homomorphism from an object to itself.

*Example.* The **identity map** $\mathrm{id}_X$ is a trivial automorphism defined

$$\mathrm{id}_X(x) = x \qquad \forall x \in X.$$

*Example.* An automorphism of a set is called a **permutation**.

### 1.1.3. Equivalance

**Definition 1.8 (Equivalance Relation)**

An **equivalance relation** on a set $A$ is a relation $R \subseteq A \times A$ with the following properties:
(i) **reflexive** $(a, a) \in R \qquad \forall a \in A$
(ii) **symmetric** $(a, b) \in R \implies (b, a) \in R$
(iii) **transitive** $(a, b), (b, c) \in R \implies (a, c) \in R$

If $(x, y) \in R$ we write $xRy$ or $x \sim y$.

Let $R$ be an equivalence relation on a set $A$.

**Definition 1.9 (Equivalence Class)**

The **equivalence class** of an element $a \in A$

$$[a] = \bar{a} = \{b \in A \mid b \sim a\}$$

is the set of all elements related to it.

*Example.* The modulo $m$ congruence class $[a] = \{a + km \mid k \in \mathbb{Z}\}$.

**Definition 1.10 (Quotient Set)**

The **quotient set** of $A$ by $R$

$$A/R = \{[a] \mid a \in A\}$$

is the set of all equivalence classes induced by the relation.

**Definition 1.11 (Partition)**

A **partition** $\mathscr{P}$ of a set $A$ is a collection of non-empty, disjoint subsets of $A$ whose union is $A$.

**Theorem 1.2**

The quotient set $A/R$ forms a partition of $A$.

**Definition 1.12 (Quotient Map)**

A **quotient map** is a map $\pi : A \to A/R$ defined by

$$\pi(a) = [a].$$

**Theorem 1.3**

Quotient maps are surjective.

## 1.2. ℤ

### 1.2.1. Linear Diophantine Equations

Let $a$ and $b$ be integers with $\gcd(a, b) = d$.

**Lemma (Bézout's Identity)**

There exist integers $x$ and $y$ such that $ax + by = d$.

**Lemma (Euclid's)**

If $p$ is prime and $p \mid ab$, then $p \mid a$ or $p \mid b$.

**Theorem 1.4 (Solvability Condition)**

The linear Diophantine equation $ax + by = c$ has (infinitely many) solutions if and only if $d \mid c$.

**Theorem 1.5 (Solutions)**

The solutions, if they exist, are

$$x = x_0 + t\frac{b}{d}, \quad y = y_0 - t\frac{a}{d}, \quad \text{for } t \in \mathbb{Z}.$$

**Theorem 1.6 (Euclidean Algorithm)**

Let $a$ and $b$ be non-negative integers with $a \geq b$. Then there exist unique integers $q$ and $r$ such that

$$a = qb + r, \quad 0 \leq r < b.$$

*Example.* We apply the algorithm to compute $\gcd(252, 105)$:

$$\begin{aligned} 252 &= 2 \cdot 105 + 42 \\ 105 &= 2 \cdot 42 + 21 \\ 42 &= 2 \cdot 21 + 0 \end{aligned}$$

Hence, $\gcd(252, 105) = 21$.

*Example.* We then use our results to solve $252x + 105y = 21$.

Since $\gcd(252, 105) = 21$, we now work backwards:

$$\begin{aligned} 21 &= 105 - 2 \cdot 42 \\ &= 105 - 2 \cdot (252 - 2 \cdot 105) \\ &= 105 - 2 \cdot 252 + 4 \cdot 105 \\ &= 5 \cdot 105 - 2 \cdot 252 \\ &= -2 \cdot 252 + 5 \cdot 105 \end{aligned}$$

So a particular solution is $x_0 = -2, y_0 = 5$. The general solutions are

$$x = -2 + 5t, \quad y = 5 - 12t, \quad \text{for } t \in \mathbb{Z}.$$

# 2. Groups

## 2.1. DEFINITIONS

### 2.1.1. Groups

**Definition 2.1 (Group)**

A **group** $(G, *)$ is a nonempty set $G$ with a binary operation

$$* : G \times G \to G$$

such that for all $a, b, c \in G$, it satisfies
(i) (Closure) $a * b \in G$
(ii) (Associativity) $(a * b) * c = a * (b * c)$
(iii) (Identity) $\exists e \in G : e * a = a * e = a$
(iv) (Inverse) $\exists a^{-1} \in G : a * a^{-1} = a^{-1} * a = e$

Additionally, a group is called **abelian** or **commutative** if it is
(v) (Commutative) $a * b = b * a$

*Examples.* The **general linear group** $GL_n(\mathbb{R})$ consists of all matrices in $\mathbb{R}^{n \times n}$ with $\det \neq 0$. The **special linear group** $SL_n(\mathbb{R})$ additionally requires that $\det = 1$.

**Tip**

The proof of (iii) and (iv) can be shortened if (v) is proven.

**Note**

Defying conventional wisdom, **abelian** is usually not capitalized.

### 2.1.2. Subgroups

Let $(G, *)$ be a group.

**Definition 2.2 (Subgroup)**

A subset $H \subseteq G$ is a **subgroup** if it is itself a group under the operation inherited from $G$.

**Theorem 2.1 (Subgroup Criterion)**

A subset $H \subseteq G$ is a subgroup if and only if
(i) $H \neq \emptyset$
(ii) $a * b^{-1} \in H \quad \forall a, b \in H$

or
(i) $H \neq \emptyset$ (nonempty)
(ii) $|H| < \infty$ (finite)
(iii) $a * b \in H \quad \forall a, b \in H$ (closed)

### 2.1.3. Cyclic

**Definition 2.3 (Cyclic Subgroup)**

For any $a \in G$, the set

$$\langle a \rangle = \{a^k \mid k \in \mathbb{Z}\}$$

forms a **cyclic subgroup** of $G$.

**Theorem 2.2**

The cyclic subgroup $\langle a \rangle \leq G$ is the smallest subgroup containing $a$.

**Definition 2.4 (Generator)**

An element $a \in G$ is a **generator** of $G$ if $\langle a \rangle = G$.

**Definition 2.5 (Cyclic Group)**

We say $G$ is a **cyclic group** if it contains a **generator** of itself.

**Theorem 2.3**

All subgroups of a cyclic group are cyclic.

**Theorem 2.4**

Every cyclic group is abelian.

*Proof.* Let $g$ and $\gamma$ be elements of a cyclic group $G = \langle a \rangle$. Then there are integers $m$ and $n$ such that $g = a^m$ and $y = a^n$, so

$$g\gamma = a^m a^n = a^{m+n} = a^{n+m} = a^n a^m = \gamma g.$$

## 2.2. FAMILIES

### 2.2.1. Commutation

Let $X$ be any nonempty subset of $G$.

**Definition 2.6 (Centralizer)**

The **centralizer** of $X$ in $G$ is the set

$$C_G(X) = \{g \in G \mid gxg^{-1} = x \text{ for all } x \in X\},$$

all elements in $G$ that commute with all elements in $X$, fixing every element of $X$ under conjugation.

**Remark**

$gxg^{-1} = x \iff gx = xg$

**Note**

If $X$ is a singleton set $\{x\}$, we simply write $C_G(x)$.

**Definition 2.7 (Center)**

The **center** of $G$ is the set

$$Z(G) = C_G(G) = \{g \in G \mid g\gamma g^{-1} = \gamma \text{ for all } \gamma \in G\},$$

all elements in $G$ that commute with all elements in $G$, fixing every element of $G$ under conjugation.

**Definition 2.8 (Normalizer)**

The **normalizer** of $X$ in $G$ is the set

$$N_G(X) = \{g \in G \mid gXg^{-1} = X\},$$

all elements in $G$ that commute with $X$ as a set, fixing $X$ as a set under conjugation.

**Theorem 2.5**

$Z(G) \leq C_G(X) \leq N_G(X) \leq G$, with all equalities if $G$ is abelian.

**Theorem 2.6**

If $X$ is a subgroup, not just a subset, then $X \leq N_G(X) \leq G$.

### 2.2.2. Group Action

**Definition 2.9 (Group Action)**

A map $\cdot : G \times X \to X$ is a **(left) group action** if it satisfies:
(i) (Identity) $e \cdot x = x$ for all $x \in X$.
(ii) (Compatibility) $g \cdot (h \cdot x) = (gh) \cdot x$ for all $g, h \in G$ and $x \in X$.

Let $\cdot : G \times X \to X$ be a group action.

**Definition 2.10 (Kernel)**

The **kernel** is the set

$$\ker(\cdot) = \{g \in G \mid g \cdot x = x \text{ for all } x \in X\},$$

all elements in $G$ that fix every element of $X$ under the group action.

**Definition 2.11 (Faithful)**

A group action is **faithful** if $\ker(\cdot) = \{e\}$.

**Definition 2.12 (Stabilizer)**

The **stabilizer** of $x \in X$ is the set

$$G_x = \{g \in G \mid g \cdot x = x\},$$

all elements in $G$ that fix $x$ under the group action.

## 2.3. COSETS

### 2.3.1. Left and Right

Let $H$ be a subgroup of a group $G$.

**Definition 2.13 (Coset)**

For any $g \in G$, the **left coset** of $H$ in $G$ is

$$gH = \{g * h \mid h \in H\}.$$

Similarly, the **right coset** is

$$Hg = \{h * g \mid h \in H\}.$$

**Theorem 2.7**

The left and right cosets respectively partition $G$.

**Theorem 2.8**

There are equally many left and right cosets of $H$ in $G$.

### 2.3.2. Order

**Definition 2.14 (Order)**

The **order of a group** $|G|$ is the number of elements in $G$.

The **order of an element** $|g|$ is the smallest positive integer $n$ such that $g^n = e$. If there is no such $n$, the order is infinite.

**Theorem 2.9 (Classification)**

If $G$ is a cyclic group of order $n$, then
(i) $G \cong \mathbb{Z} \iff n = \infty$
(ii) $G \cong \mathbb{Z}_n \iff n < \infty$

**Definition 2.15 (Index)**

The **index** of $H$ in $G$ is the number of left cosets, denoted $[G : H]$.

**Theorem 2.10 (Lagrange's)**

Let $G$ be finite. Then the order of $H$ divides the order of $G$.

**Corollary**

If $G$ is of prime order, it
(i) lacks proper nontrivial subgroups
(ii) is generated by any element $g \neq e$
(iii) is cyclic

**Theorem 2.11 (Lagrange's, Generalized)**

Regardless of whether G is finite or infinite,

$$|G| = [G : H]\ |H|.$$

*Proof.* $G$ has $|G|$ elements and is partitioned into $[G : H]$ distinct left cosets, each of which has $|H|$ elements.

**Theorem 2.12**

Let $(g, h) \in G \times H$ be a tuple of groups. If they have finite orders $|g| = r < \infty$ and $|h| = s < \infty$, then $|(g, h)| = \operatorname{lcm}(r, s)$.

**Corollary**

$\mathbb{Z}_m \times \mathbb{Z}_n \cong \mathbb{Z}_{mn} \iff \gcd(m, n) = 1$

### 2.3.3. Conjugation

**Definition 2.16 (Conjugation)**

The **conjugation** by $a \in G$ is an **inner automorphism** of $G$ defined $g \mapsto gag^{-1}$.

**Definition 2.17 (Normal Subgroup)**

The following are equivalent for a subgroup $H$ of $G$:
(i) $H$ is a **normal subgroup** of $G$, denoted $H \lhd G$.

(ii) $ghg^{-1} \in H$ for all $g \in G$ and $h \in H$.
(iii) $gHg^{-1} \subseteq H$ for all $g \in G$.
(iv) $gHg^{-1} = H$ for all $g \in G$.

(v) $gH = Hg$ (equal left coset and right coset) for all $g \in G$.
(vi) The sets of left and right cosets coincide.

*Examples.* Trivially, $G$ and $\{e\}$ are always normal subgroups.

**Theorem 2.13**

Every subgroup of index 2 is normal.

*Proof.* Let $H \leq G$ be the subgroup such that $[G : H] = 2$. We wish to show that $gH = Hg$ for all $g \in G$.

If $g \in H$, then $gH = H = Hg$. Otherwise, if $g \notin H$, then $gH = G/H = Hg$ since $H \cup Hg = G = H \cup gH$.

Thus, $gH = Hg$ for all $g$, so $H$ is a normal subgroup of $G$.

**Definition 2.18 (Simple Group)**

A group is **simple** if it has no normal nontrivial proper subgroup.

**Theorem 2.14**

All subgroups of abelian groups are normal subgroups.

### 2.3.4. The Isomorphism Theorems

Let $\varphi : G \to H$ be a group homomorphism.

**Theorem 2.15 (First or Fundamental of Homomorphisms)**

$\ker \varphi \trianglelefteq G$ and $G/\ker \varphi \cong \varphi(G)$

**Corollary**

(i) $\varphi$ is injective if and only if $\ker \varphi = \mathrm{id}_G$
(ii) $|G : \ker \varphi| = |\varphi(G)|$

**Theorem 2.16 (Second or Diamond)**

Let $A \leq G$ and $B \leq G$ be subgroups such that $A \leq N_G(B)$. Then,
(i) $AB \leq G$
(ii) $B \trianglelefteq AB$
(iii) $A \cap B \trianglelefteq A$
(iv) $AB/B \cong A/A \cap B$

## 2.4. RINGS

### 2.4.1. Definition

**Definition 2.19 (Ring)**

A **ring** is a set $R$ with two binary operations $+$ and $\cdot$ such that
(i) $(R, +)$ is an abelian group with identity 0.
(ii) $(R, \cdot)$ is a monoid with identity 1.
(iii) Multiplication distributes over addition:
for all $a, b, c \in R$, $a \cdot (b + c) = a \cdot b + a \cdot c$.

**Definition 2.20 (Group of Units)**

Let $(R, +, \cdot)$ be a ring. The **group of units** $U(R)$ is the set of all elements in $R$ that have multiplicative inverses, that is

$$U(R) = \{a \in R \mid \exists a^{-1} \in R : a \cdot a^{-1} = a^{-1} \cdot a = 1\}.$$

### 2.4.2. $\mathbb{Z}_n$

Let $a$ and $n$ be positive integers and let $p$ be prime.

**Definition 2.21 ($\mathbb{Z}_n$)**

The **integers modulo $n$** is a set of congruence classes

$$\mathbb{Z}_n = \{[0], [1], ..., [n-1]\}$$

that form an abelian group under $+$ and a monoid under $\times$, together forming a ring with a group of units called the **multiplicative group of integers modulo $n$**, denoted $U(n)$ and consisting of the integers coprime to $n$, with inverse elements defined to satisfy $aa^{-1} (\equiv a^{-1}a) \equiv 1 \pmod{n}$.

**Definition 2.22 (Euler's Totient Function)**

$$\phi(n) = |U(n)| = |\{k \in \mathbb{Z} \mid 1 \leq k \leq n \text{ and } \gcd(n, k) = 1\}|$$

*Example.* $\phi(12) = |\{1, 5, 7, 11\}| = 4$

**Theorem 2.17 (Euler's)**

If $a$ and $n$ are coprime, then

$$a^{\phi(n)} \equiv 1 \pmod{n}.$$

*Proof.* The order of $a$ is $|U(n)| = \phi(n)$.

**Lemma**

$\phi(p) = p - 1$

**Theorem 2.18 (Fermat's Little)**

If $a$ and $p$ are coprime, then

$$a^{p-1} \equiv 1 \pmod{p}.$$

In general, even if they are not coprime,

$$a^p \equiv a \pmod{p}.$$

## 2.5. PERMUTATIONS

### 2.5.1. Transpositions

**Definition 2.23 (Transposition)**

A **transposition** of the elements $i, j \in \{1, ..., n\}$ is the permutation $\tau_{ij}$ with the cycle notation $(i\ j)$.

**Theorem 2.19**

Any permutation can be written as compositions of transpositions.

**Theorem 2.20**

The identity permutation may only be written as an even number of transpositions.

**Definition 2.24 (Sign of a Permutation)**

Let $k$ be the number of transpositions in any decomposition of a permutation $\sigma \in S_n$. The **sign** of $\sigma$ is $(-1)^k$.

Hence,

- if $k$ is **even** then $\operatorname{sgn}(\sigma) = +1$ and $\sigma$ is said to be even.
- if $k$ is **odd** then $\operatorname{sgn}(\sigma) = -1$ and $\sigma$ is said to be odd.

**Theorem 2.21 (Permutation Inverse)**

Let $\sigma \in S_n$ be a permutation.
(i) $\sigma = (a_1\ a_2\ \cdots\ a_m) = \tau_1 \tau_2 \cdots \tau_k$ for some $m \leq n$ and $k$
(ii) $\sigma^{-1}$ exists

*Proof.* $\sigma$ is a bijection.

(iii) $\sigma^{-1} = (a_m\ \cdots\ a_2\ a_1) = \tau_1 \tau_2 \cdots \tau_k$

*Proof.* Consider that $(\tau_i \tau_j)^{-1} = \tau_j^{-1} \tau_i^{-1} = \tau_j \tau_i$ and $(a\ b\ c)^{-1} = (a \to b \to c \to a)^{-1} = (a \leftarrow b \leftarrow c \leftarrow a)$.

(iv) $\operatorname{sgn}(\sigma^{-1}) = \operatorname{sgn}(\sigma) = (-1)^k$

**Definition 2.25 (Fixed Point))**

A **fixed point** is invariant under a transformation such as a permutation.

### 2.5.2. Symmetric Groups

**Definition 2.26 (Symmetric Group)**

The **symmetric group** of the finite set $X = \{1, 2, ..., n\}$ is

$$S_n = \{\sigma : X \to X \mid \sigma \text{ is bijective}\}$$

and consists of $n!$ permutations, which form **the symmetric group on $n$ letters** under the operation of composition

**Theorem 2.22**

The automorphism group of a set is its symmetric group.

**Definition 2.27 (Permutation Group)**

A subgroup of a symmetric group $S_n$ is a **permutation group**.

**Definition 2.28 (Alternating Group)**

The **alternating group**

$$A_n = \{\sigma \in S_n \mid \operatorname{sgn}(\sigma) = 1\}$$

is the subgroup of $S_n$ consisting of all even permutations.

It has $n!/2$ elements for $n \geq 2$.

*Problem.* Show that $A_4$ has no subgroup of order 6.

*Solution.*

**Theorem 2.23**

The sign map $\operatorname{sgn} : S_n \to \{1, -1\}$ is a group homomorphism.

The kernel of this homomorphism is the alternating group

$$\ker(\operatorname{sgn}) = A_n.$$

## 2.6. SYMMETRY

### 2.6.1. In Metric Spaces

**Definition 2.29 (Isometry)**

Let $(X, d_X)$ and $(Y, d_Y)$ be metric spaces. A map $f : X \to Y$ is an **isometry** if it perserves distances, that is,

$$d_X(x_1, x_2) = d_Y(f(x_1), f(x_2)) \quad \forall x_1, x_2 \in X.$$

**Definition 2.30 (Symmetry)**

A **symmetry** of an object $O \subseteq X$ in a metric space $(X, d)$ is an isometry $f : X \to X$ with the image $f(O) = O$.

Informally, a symmetry of a geometric object is a rearrangment of the object preserving the arrangement of its sides and vertices as well as its distances and angles.

*Examples.* Some types of symmetries:
(i) **Reflectional** Reflection across a hyperplane (a line in $\mathbb{R}^2$ or a plane in $\mathbb{R}^3$)
(ii) **Rotational** Rotation within a plane ($\mathbb{R}^2$ is rotated around a fixed point, while $\mathbb{R}^3$ is rotated around an axis).
(iii) **Translational** All points are shifted by a constant vector.
(iv) **Identity** No change.

**Definition 2.31 (Rigid Motion)**

A **rigid motion** is a map $f : \mathbb{R}^n \to \mathbb{R}^n$ such that

$$\|u - v\| = \|f(u) - f(v)\| \quad \forall u, v \in \mathbb{R}^n.$$

Equivalently, it is a surjective isometry between Euclidean spaces

*Example.* A symmetry of an object in an Euclidean space $X = \mathbb{R}^n$ is also a rigid motion.

### 2.6.2. Dihedral Groups

**Definition 2.32 (Dihedral Group)**

The **dihedral group** $D_n$ is the group of symmetries (or equivalently rigid motions) of an $n$-gon. Its degree is $n$ and its order is $2n$.

**Theorem 2.24**

The dihedral group can be represented as

$$D_n = \langle r, s \mid r^n = s^2 = 1, rs = sr^{-1} \rangle.$$

# 3. Fields

## 3.1. DEFINITION

**Definition 3.1 (Field)**

A **field** is a set $F$ with two binary operations $+$ and $\cdot$ such that
(i) $(F, +)$ is an abelian group with identity 0.
(ii) $(F \setminus \{0\}, \cdot)$ is an abelian group with identity 1.
(iii) Multiplication distributes over addition:
for all $a, b, c \in F$, $a \cdot (b + c) = a \cdot b + a \cdot c$.

*Examples.* $\mathbb{Q}, \mathbb{R}, \mathbb{C}, \mathbb{R}(x), \mathbb{Q}(i)$, and $\mathbb{F}_p$ for prime $p$.

## 3.2. VECTOR SPACES

**Definition 3.2 (Vector Space)**

A vector space over a field $F$ is a set $V$ with two operations:

- Vector addition $+ : V \times V \to V$
- Scalar multiplication $\cdot : F \times V \to V$

These must satisfy
(i) (Closure under addition) $\mathbf{u} + \mathbf{v} \in V$
(ii) (Distributivity over vectors) $a(\mathbf{u} + \mathbf{v}) = a\mathbf{u} + a\mathbf{v}$
(iii) (Associativity) $\mathbf{u} + (\mathbf{v} + \mathbf{w}) = (\mathbf{u} + \mathbf{v}) + \mathbf{w}$
(iv) (Identity) $\mathbf{u} + \mathbf{0} = \mathbf{u}$

(v) (Inverse) $\mathbf{u} + (-\mathbf{u}) = \mathbf{0}$
(vi) (Commutativity) $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$

(vii) (Closure under scalar multiplication) $a\mathbf{u} \in V$
(viii) (Distributivity over scalars) $(a + b)\mathbf{u} = a\mathbf{u} + b\mathbf{u}$
(ix) (Associativity) $a(b\mathbf{u}) = (ab)\mathbf{u}$
(x) (Identity) $1\mathbf{u} = \mathbf{u}$

Note that requiring closure is often omitted.